Balance Training With a Dynamometric Platform Following Total Knee Replacement: A Randomized Controlled Trial

Sergio Roig-Casasús, PT, PhD[1,2]; José María Blasco, PhD[1]; Laura López-Bueno, PT, PhD[1,2]; María Clara Blasco-Igual, PhD[1]

ABSTRACT

Background and Purpose: Sensorimotor training has proven to be an efficient approach for recovering balance control following total knee replacement (TKR). The purpose of this trial was to evaluate the influence of specific balance-targeted training using a dynamometric platform on the overall state of balance in older adults undergoing TKR.

Methods: This was a randomized controlled clinical trial conducted at a university hospital rehabilitation unit. Patients meeting the inclusion criteria were randomly assigned to a control group or an experimental group. Both groups participated in the same 4-week postoperative rehabilitation training protocol. Participants in the experimental group performed additional balance training with a dynamometric platform consisting of tests related to stability challenges, weight-shifting, and moving to the limits of stability. The primary outcome measure was the overall state of balance rated according to the Berg Balance Scale. Secondary outcomes in terms of balance were the Timed Up and Go Test, Functional Reach Test, and Romberg open and closed-eyes tests. Data processing included between-group analysis of covariance, minimal detectable change assessment for the primary outcome measure, and effect size estimation. Confidence intervals (CIs) were set at 95%.

Results: Forty-three participants meeting the inclusion criteria and having signed the informed consent were randomly assigned to 2 groups. Thirty-seven completed the training (86.1%). Significant between-group differences in balance performance were found as measured with the Berg Balance Scale ($P = .03$) and Functional Reach Test ($P = .04$) with a CI = 95%. Significant differences were not recorded for the Timed Up and Go Test or Romberg open and closed-eyes tests ($P > .05$). Furthermore, Cohen's effect size resulted in a value of $d = 0.97$, suggesting a high practical significance of the trial.

Discussion and Conclusions: According to the Berg Balance Scale and Functional Reach Test, participants with TKR who have followed a 4-week training program using a dynamometric platform improved balance performance to a higher extent than a control group training without such a device. The inclusion of this instrument in the functional training protocol may be beneficial for recovering balance following TKR.

Key Words: advanced knee osteoarthrosis, balance, dynamometric platform, proprioception, total knee replacement

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INTRODUCTION

Symptoms of knee osteoarthritis include joint pain, stiffness, limited mobility, functional deficits, and proprioceptive impairements.[1] When conservative treatments fail to effectively control pain, total knee replacement (TKR) is a common surgical procedure used to reduce symptoms due to severe osteoarthritis. After surgery, pain and stiffness are remarkably relieved, but functional and proprioceptive deficits, including limitation of lower limb mobility, difficulties in walking, and alterations in stability and balance,[2,3] may remain.[4]

Rehabilitation following TKR has mainly been focused on restoring joint function and strength, and to a lesser extent on improving balance and proprioception.[5] However, several authors have suggested that rehabilitation methods might ideally include task-oriented protocols focused on balance recovery.[6-9] Indeed, sensorimotor training has been reported to be an efficient approach to balance recovery following TKR.[10] Therefore, this research especially focuses on evaluating the effect of specific training to improve balance after TKR.

The use of a dynamometric platform is proposed as an instrument to be included among the methods for recovering and enhancing balance. Traditionally, dynamometric platforms have been used for patients with vestibular pathology,[11] whereas the use in the area of physical rehabilitation is still under research, including for patients with osteoarticular and musculotendinous pathologies.[12]

[1]*Department of Physiotherapy, Faculty of Physiotherapy, University of Valencia, Valencia, Spain.*

[2]*Hospital Clínic Universitari de València, Valencia, Spain.*

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Address correspondence to: José María Blasco, PhD, Department of Physiotherapy, Faculty of Physiotherapy, University of Valencia, Calle Gascó Oliag nº 5, 46010, Valencia, Spain (jose.maria.blasco@uv.es)

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This research is aimed at determining whether a training protocol including the use of a dynamometric platform might be beneficial in the restoration of balance in patients undergoing TKR when compared with a traditional functional rehabilitation protocol.

METHODS

Design, Setting, and Participants

This was a prospective, randomized 2-arm clinical trial focusing on rehabilitation subsequent to hospitalization for TKR. The research was performed between March 2009 and December 2012. The trial was approved by the Scientific and the Ethics Committee for Clinical Research of the Clinical University Hospital of Valencia. The procedures followed were in accordance with the ethical standards pursuant to the Declaration of Helsinki, as revised in Hong Kong.

Considerations for inclusion required that patients had advanced knee osteoarthritis for which a first TKR was performed, were older than 65 years, scored between 21 and 56 on the Berg Balance Scale[13] indicating a medium-low risk of falling and scored between 20 and 30 on the Mini-Mental State Examination[14] indicating no severe cognitive impairment, were fitted with the same Vanguard Complete Knee System, and were willing to participate in the trial, understanding it and agreeing to sign an informed consent. Exclusion criteria applied to those patients with morphological alterations of the hip or ankle, suspected deep vein thrombosis, postoperative infection of the operated knee, and either central (ie, cerebellar) or vestibular pathology. Criteria were initially administered by the orthopedic surgeon based on the medical chart, whereas the other necessary physical examinations were administered by the physiotherapist.

Patients meeting the inclusion criteria were asked to participate. Each participant was allocated according to a random number table extracted from a software tool (Matlab, The MathWorks, Inc, Natick, Massachusetts). The rehabilitation physician who forwarded the participants to the physiotherapist was blinded to the training allocation. Participants were informed of the sessions by phone. The physiotherapist in charge of the training was not blinded to the groups. A person with no knowledge of the participant's allocation was designated to extract and interpret the results. Data were collected at the hospital facilities.

Training

The physiotherapy training aimed to achieve short-term functional and balance recovery. Baseline was 2 weeks after surgery. The training lasted 4 weeks, with a total of 20 sessions administered at a university hospital. Follow-up was 6 weeks after intervention. Following surgery, a control group (CG) and an experimental group (EG) participated in the same training protocol, whereas training with a dynamometric platform was included for the EG.

The CG followed rehabilitation training based on the following aspects: functional exercises, muscle strengthening with isometric, isotonic, and counter-resistance exercises, as well as balance and proprioception exercises. The training protocol described in Appendix 1 consisted of a warm-up phase with passive, active-assisted, and active movements, functional and balance task-oriented exercises and cool-down activities; it was based on Brotzman's protocol[15] and clinical experience. Each session lasted approximately 60 minutes.

The EG performed the same training, which additionally included exercises with a dynamometric platform based on stability challenges, weight-shifting, and moving to the limits of stability. The platform was the Biodex Balance System SD (BBS). The participants were instructed to keep hands parallel to the body and to maintain a comfortable knee angle, and were also asked to stand centered on the platform in a position enabling good stability.[16] For safety reasons and only to the extent required to correct the posture in extreme positions, participants were allowed to touch the handrails. These were previously adjusted for each participant. Foot position was set with feet at an angle of 20º. The participants were allowed to rest between trials and exercises in order to reduce fatigue. Additional training time was approximately 20 minutes. Detailed information on the additional EG platform training protocol is presented in Appendix 2.

Outcome Measures

Validated tests and scales for evaluating balance were included.[17] The primary outcome measure rating the overall state of balance for older adults was the Balance Berg Scale.[13] Secondary balance outcome measures using the Timed Up and Go Test,[18] the Functional Reach Test,[19] and the Romberg open and closed-eyes test. The measures were obtained at baseline and after training.

The Berg Balance Scale has shown a high degree of inter and intraobserver reliability and validity, offering a good correlation with other clinical assessment scales.[20] It is a 14-item balance assessment tool, in which each item is scored on a 0 to 4 scale, the total maximum being 56.

The Timed Up and Go Test, used to assess dynamic balance, is a test of general mobility, in which the individual rises from a standard arm chair without using their arms, walks 3 m, turns around, returns to the chair, and sits down. The test score measured in seconds was the mean of 2 trials.

The Functional Reach Test, used to detect balance deterioration and changes in performance over time, measured the maximum distance the participant could reach forward while standing in a fixed position. Procedures were adapted based on those described by Duncan et al.[19] The participant was asked to stand close to a wall, without touching it, with 90º of shoulder flexion; the participant was then instructed to reach as far as possible without taking a step. The difference between the start and end position was measured in centimeters. Three trials were conducted and the mean value of the last 2 was noted.

The Romberg test, used to assess static balance and proprioception, determines the participant's ability to maintain the center of balance. BBS testing protocol consisted of 3 consecutive 20-second trials, with 2-leg stance.

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The tests were conducted with open and closed eyes. The participant's performance was noted as the mean value of the overall stability index provided by the BBS. This index represents the variance of platform displacement in degrees from level. The score assesses deviations from center, thus a lower score is more desirable than a higher score.

Data Analysis

Data were analyzed with the Matlab software tool (Statistics Toolbox, Release 2013) and have been expressed as the mean result of each variable, both at baseline and after training, as well as the standard deviation. Between-group differences were estimated with the analysis of covariance (ANCOVA) to assess whether the mean score for each outcome measured at the end of the training and adjusted for baseline scores, differed between the 2 groups. Confidence intervals to state between-group significant differences were set at 95%.

To test whether likely differences were clinically important, the minimal detectable change (MDC) was compared against the between-group mean change. This information was extracted from the literature. For the primary outcome measure, the Berg Balance Scale, an $MDC_{95\%}$ of 6.5 points was established by Romero et al.[21] For the secondary outcomes, the Timed Up and Go Test, Functional Reach Test, or Romberg tests, MDCs have been reported to be considerably variable, depending on the pathology, for instance Alzheimer's disease, stroke, or Parkinson's disease. To our knowledge, none of these matched or were similar to the conditions of the population under study and were not considered.

The G*Power software tool[22] was proposed to estimate the clinically important difference of the trial by using a distribution-based method based on the effect size calculated from the sample size and the primary outcome measure results.

RESULTS

A total of 69 patients who had undergone surgery for advanced knee osteoarthritis were recruited as potential participants. Those meeting the inclusion criteria took part in the randomization process. Thirty-seven participants from both groups attended a mean of 19.4 (0.8) sessions ranging from 18 to 20 sessions. There was a nonsignificant difference in compliance between the 2 rehabilitation groups. Six of the participants (13.9% of the total sample) did not complete training. Data pertaining to the latter were available at baseline, but not in the follow-up assessment. To our knowledge, there were no adverse events or clinical complications related to the trial. The flow diagram is shown in the Figure, whereas the demographic data appear in Table 1.

The assessment of both groups at baseline, after training, and the progression and significance levels of the between-group comparison are shown in Table 2.

Nonsignificant between-group differences were found at baseline in all of the outcome measures: Timed Up and Go Test ($P = .10$), Berg Balance Scale ($P = .10$), Functional Reach Test ($P = 1.00$), Romberg open-eyes ($P = .28$), and Romberg closed-eyes ($P = .79$).

Between-group comparison showed significant differences in terms of the Berg Balance Scale and the Functional Reach Test in favor of the experimental group ($P < .05$) when controlling for baseline status with ANCOVA. A mean change in the Berg Balance Scale of 6.8 points was likely relevant as it was over the 6.5 points established for

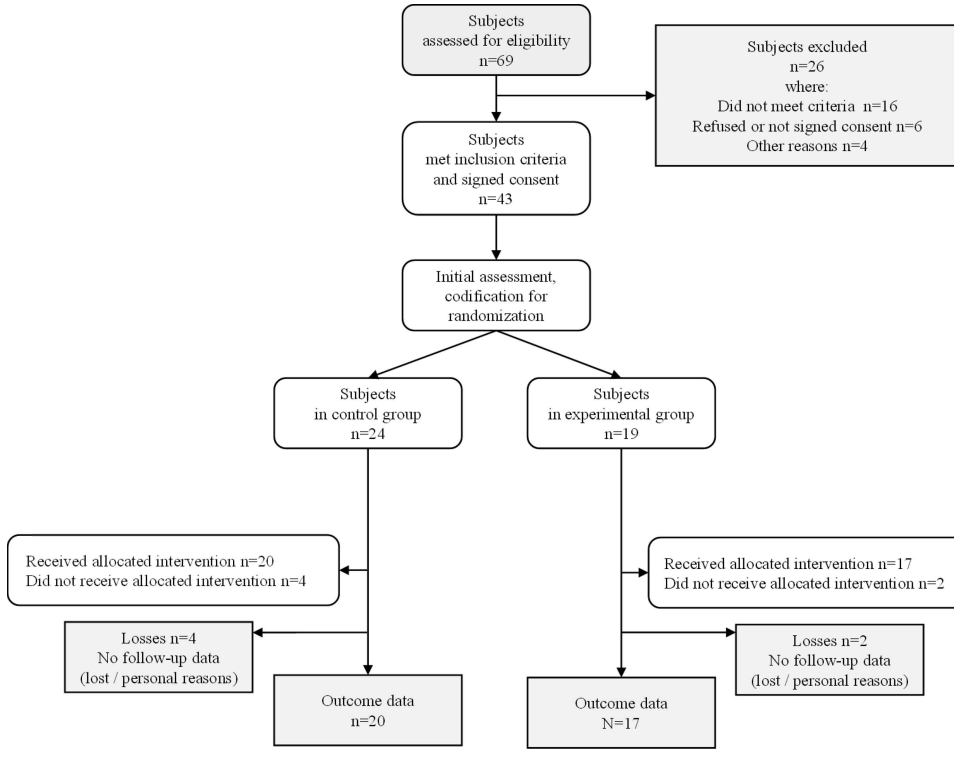


Figure. A flow chart describing the patients of both groups.

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Table 1. Demographic Data of the Participants

Sample Descriptor	Control Group	Experimental Group	Total
Age, y	72.1 (4.5)	74.8 (4.0)	73.4 (4.4)
Weight, kg	72.9 (5.9)	69.5 (6.4)	71.2 (6.3)
Gender			
Male	7	5	12
Female	13	12	25
Operated knee			
Right	8	8	16
Left	12	9	21

Summary data are presented as n or mean (SD).

the $MDC_{95\%}$.[21] Qualitative visual observation of the Timed Up and Go Test and Romberg open-eyes mean differences showed a higher improvement in the EG, although differences were nonsignificant. Matched performances were observed for the Romberg closed-eyes outcome, also with nonsignificant differences.

Power analysis[22] for the Berg Balance Scale as the main outcome by using the final sample size of the trial resulted in a power of 0.83. Furthermore, Cohen's effect size resulted in a value of $d = 0.97$.

DISCUSSION

The main finding was that a 4-week training program whose methods included a dynamometric platform resulted in significant improvements of the EG as measured with the primary outcome measure, the Berg Balance Scale, and the secondary outcome measure, and the Functional Reach Test, when compared against the CG. This supports the belief that specific balance training with a dynamometric platform in combination with functional training may be a suitable technique to recover balance after TKR.

In the early TKR postoperative phase (baseline), the measurements showed that the participants' balance parameters were below the levels considered normal for community-dwelling elderly people, indicating that both groups presented instability and a high risk of falling; for instance the scores in the Timed Up and Go Test were over the cut-off point of 13.5 seconds for the risk of falling proposed by Shumway-Cook et al,[23] below 45 on the Berg Balance Scale,[13,24] and below 27 cm in the Functional Reach Test.[25] Numerous authors[6-9,26] have stressed the importance of restoring balance and focused efforts on evaluating the effectiveness of functional plus balance training over traditional functional training in patients undergoing TKR, and reported satisfactory results. However, the usefulness of dynamometric platforms as a training tool for balance recovery following TKR had not been evaluated. Because these are common instruments among many biomedical specialists, and in recent years have been progressively introduced into many research facilities and hospitals, further studying their potential was one of the goals pursued in this research.

Focusing on posttraining findings, the score recorded on the Berg Balance Scale was an indicator that participants achieved the ability to walk autonomously and, accordingly, the risk of falling was reduced, as the average test values were 52 out of a maximum of 56.[13] In this regard, the results are consistent with balance recovery-oriented rehabilitation protocols, such as Jogi et al's,[6] which quantified the same outcome for balance assessment after TKR. In addition, the between-group mean difference of 6.8 points was above the established $MDC_{95\%}$ threshold of 6.5 points,[21] which suggests a meaningful clinical difference due to the additional EG training. In short, functional training combined with dynamometric platform exercises produced a good overall balance recovery in the early postoperative stage.

For the baseline Timed Up and Go Test assessment, times between 20 and 30 seconds were recorded. After rehabilitation, a considerable decrease was registered, and EG participants showed the largest relative progress, although between-group differences were nonsignificant. Times were above normal limits for community-dwelling older adults[27] and did not reach the aforementioned Shumway-Cook et al's[23] threshold (see Table 2). By contrast, Liao et al's[9] functional plus balance training reported shorter times; however, the baseline was set at a later stage, and the protocol involved a more extensive exercise proposal, which might bias this comparison. A greater connection was found with Jogi et al's[6] research with a similar timeline. Hence, this fact may presumably be due to the need for a longer recovery time to optimize this result.[4]

The Romberg open-eyes test showed greater improvements (nonsignificant) for the EG when compared with the CG. By contrast, the improvements achieved in the closed-

Table 2. Outcome Results and Analysis of Covariance Between-Group Analyses[a]

Outcome	Baseline		End of Intervention		Change	*P* Value
	Control	Experimental	Control	Experimental		
Berg Balance Scale, score	34.6 (5.6)	29.9 (6.8)	49.7 (3.8)	51.8 (2.7)	6.8 (9.9)	.03[b]
Timed Up & Go Test, s	25.1 (7.5)	25.2 (4.5)	17.3 (3.6)	14.4 (4.3)	3.0 (10.3)	.13[c]
Romberg open-eyes, score	0.74 (0.32)	0.71 (0.40)	0.60 (0.20)	0.24 (0.33)	0.33 (0.64)	.09[c]
Romberg closed-eyes, score	1.97 (0.33)	1.64 (0.56)	1.51 (0.34)	1.12 (0.35)	0.06 (0.81)	.07[c]
Functional Reach Test, cm	18.4 (5.6)	17.1 (6.5)	24.4 (5.9)	27.7 (6.7)	4.6 (12.3)	.04[b]

[a]Data are given as mean (standard deviation). Romberg scores are presented as the Biodex overall stability index.
[b]$P < .05$.
[c]Nonsignificant.

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eyes test were very similar (also nonsignificant differences). A positive result indicates a deficit in the proprioceptive pathway, which may be located at any point,[28] and therefore, results suggest that treatment in the EG, despite the improvements, did not result in further recovery of static balance with closed eyes. The recovery of lower limb proprioception and of the static equilibrium seems to take longer than the dynamic balance recovery, as pointed out by Guachard et al.[7]

According to the Functional Reach Test, specific training with the platform conducting tests of stability challenges, weight-shifting, and moving to the limits of stability, along with simultaneous anteroposterior and medial-lateral movements, had a significant impact on the participants' recovery. Results are consistent with those obtained by Liao et al[9] for this specific outcome. Although a corrected value was given in that research, when balance performances were estimated from the Functional Reach Test, a significant improvement was demonstrated in the EG when compared to the CG.

In summary, findings suggested potential clinical benefits of specific balance training following TKR, and supported current evidence[6-9,26] implying its feasibility and likely efficacy. The clinical importance based on the effect size estimation led to a high practical significance of the trial. In addition, the clinical relevance of this trial suggests that a dynamometric platform may be a useful device to recover balance as part of a functional rehabilitation training.

Importantly, there are several limitations to this research. Participant unavailability for the follow-up assessment may be a source of bias. Early postoperative outcomes have been reported to be satisfactory; however, it can take from 6 months to 1 year to reach the functional plateau values after a TKR surgery.[4] The lack of this follow-up information is one of the main limitations of this trial, whereas the role in the long-term improvement due to the effect of the added training for the EG may require further research. A post hoc analysis[22] for the primary outcome and the sample size of the trial resulted in a power of 0.83. There is much debate about the use of post hoc analyses, so the estimation should preferably be conducted a priori. Higher scores achieved in the open-eyes Romberg test might be influenced by the specific platform training practiced by the EG, bearing in mind that the CG participants were less familiar with the instrument. By contrast, being somewhat familiar with the platform did not affect the closed-eyes test results, which could preclude the above assertion.

CONCLUSIONS

This research demonstrated that a 4-week functional training protocol that included a dynamometric platform in the training methods enhanced balance performance of older adults in the early TKR postoperative stage to a higher extent than a traditional functional training protocol. The statement is supported by the primary outcome measure, the Berg Balance Scale, as well as the Functional Reach Test between-group results. Research findings substantiate previous evidence pointing to the efficacy and potential clinical importance of specific balance training following TKR, and suggest the usefulness of a dynamometric platform to recover balance as part of a training method in combination with a functional rehabilitation program.

REFERENCES

1. Taş S, Güneri S, Baki A, Yildirim T, Kaymak B, Erden Z. Effects of severity of osteoarthritis on the temporospatial gait parameters in patients with knee osteoarthritis. *Acta Orthop Traumatol Turc*. 2014;48(6):635-641.
2. Viton JM, Atlani L, Mesure S, et al. Reorganization of equilibrium and movement control strategies after total knee arthroplasty. *J Rehabil Med*. 2002;34(1):12-19.
3. Rossi MD, Hasson S, Kohia M, Pineda E, Bryan W. Mobility and perceived function after total knee arthroplasty. *J Arthroplasty*. 2006;21(1):6-12.
4. Bade MJ, Kohrt WM, Stevens-Lapsey JE. Outcomes before and after total knee arthroplasty compared to healthy adults. *J Orthop Sports Phys Ther*. 2010;40(9):559-576.
5. Minns Lowe CJ, Barker KL, Dewey M, Sackley CM. Effectiveness of physiotherapy exercise after knee arthroplasty for osteoarthritis: systematic review and meta-analysis of randomised controlled trials. *BMJ*. 2007;335(7624):1-9.
6. Jogi P, Overend TJ, Spaulding SJ, Zecevic A, Kramer JF. Effectiveness of balance exercise programs following total hip and knee joint arthroplasty. *SAGE Open Med*. 2015;3:2050312115570769.
7. Guachard GC, Vançon G, Meyer P, Mainard D, Perrin PP. On the role of knee joint in balance control and postural strategies: effects of total knee replacement in elderly subjects with knee osteoarthritis. *Gait Posture*. 2010;32(2):155-160.
8. Piva SR, Gil AB, Almeida GJ, DiGioia AM, Levison TJ, Fitgerald GK. A balance exercise program appears to improve function for patients with total knee arthroplasty: a randomized clinical trial. *Phys Ther*. 2010;90(6):880-894.
9. Liao CD, Liou TH, Huang YY, Huang YC. Effects of balance training on functional outcome after total knee replacement in patients with knee osteoarthritis: a randomized controlled trial. *Clin Rehabil*. 2013;27(8):697-709.
10. Moutzuri M, Gleeson N, Billis E, Panoutsopolou I, Gliatis J. What is the effect of sensori-motor training on functional outcome and balance performance of patients' undergoing TKR? A systematic review. *Physiotherapy*. 2016;102(2):136-144.
11. Furman JM. Posturography: uses and limitations. *Baillieres Clin Neurol*. 1994;3(3):501-513.
12. Peydro MF, Baydal BJM, Vivas BMJ. Assessment and rehabilitation of balance by posturography. *Rehabilitación*. 2005;39(6):315-323.
13. Berg KO, Wood-Dauphine SL, Williams JL, Maki B. Measuring balance in the elderly: validation of an instrument. *Can J Public Health*. 1992;83(2):7-11.
14. Folstein MF, Robins NL, Helzer JE. The Mini-Mental State Examination. *Arch Gen Psychiatry*. 1983;40(7):812.
15. Brotzman SB, Wilk KE. Artrosis de la extremidad inferior. In: *Rehabilitación Ortopédica Clínica, ed*. Mosby; 2005:50-458.
16. Arnold BL, Schmitz RJ. Examination of balance measures produced by the Biodex stability system. *J Athl Train*. 1998;33(4):323-327.
17. Mancini M, Horak FB. The relevance of clinical balance assessment tools to differentiate balance deficits. *Eur J Phys Rehabil Med*. 2010;46(2):239-248.
18. Podsialo D, Richardson S. The timed "up and go": a test of basic functional mobility for frail elderly persons. *J Am Geriatr Soc*. 1991;39(2):142-148.
19. Duncan PW, Weiner DK, Chandler J, Studenski S. Functional reach: a new clinical measure of balance. *J Gerontol*. 1990;45(6):192-197.
20. Berg KO, Maki BE, Williams JI, Holliday PJ, Wood-Dauphinee SL. Clinical and laboratory measures of postural balance in an elderly population. *Arch Phys Med Rehab*. 1992;73(11):1073-1080.
21. Romero S, Bishop MD, Velozo CA, Light K. Minimum detectable change of the Berg Balance Scale and Dynamic Gait Index in older persons at risk for falling. *J Geriatr Phys Ther*. 2011;34(3):131-137.
22. Faul F, Erdfelder E, Lang AG, Buchner A. G*Power 3: a flexible statistical power analysis program for the social, behavioral, and biomedical sciences. *Behav Res Methods*. 2007;39(2):175-191.
23. Shumway-Cook A, Brauer S, Woollacott M. Predicting the probability for falls in community-dwelling older adults using the timed up & go test. *Phys Ther*. 2000;80(9):896-903.
24. Muir SW, Berg K, Chesworth B, Speechley M. Use of the Berg Balance Scale for predicting multiple falls in community-dwelling elderly people: a prospective study. *Phys Ther*. 2008;88(4):449-459.
25. Goldberg A, Hernandez ME, Alexander NB. Trunk repositioning errors are increased in balance-impaired older adults. *J Gerontol A Biol Sci Med Sci*. 2005;60(10):1310-1314.
26. Liao CD, Lin LF, Huang YC, Huang SW, Chou LC, Liou TH. Functional outcomes of outpatient balance training following total knee replacement in patients with knee osteoarthritis: a randomized controlled trial. *Clin Rehabil*. 2015;29(9):855-867.
27. Bohannon RW. Reference values for the timed up and go test: a descriptive meta-analysis. *J Geriatr Phys Ther*. 2006;29(2):64-68.
28. Pearce JM. Romberg and his sign. *Eur Neurol*. 2005;53(4):210-213.

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Appendix 1. Functional Training Program

Phase	Exercise	Description	Objective	Dose
Warm-up	Ankle pumps	Patient in the long sitting position performs ankle dorsiflexion and plantar flexion	Muscle flexibility, joint mobility, warm-up	3 min
	Stretching quadriceps and hamstrings	Patient in long sitting position. The nonoperated leg over the table with 90º hip and knee flexion. For stretching quadriceps, the operated leg hangs off the table. For hamstrings stretching the operated leg in extension along the table	Muscle flexibility, joint mobility, warm-up	Stretching 3 sets of 30 s each muscle group
	Knee flexion and extension mobilization exercises	Patient in the sitting position. Passive, active-assisted or active movements are performed according to patient progression	Muscle flexibility, joint mobility, warm-up	3 min
Strength	Isometric quadriceps	Patient in the sitting position and knee extension	Muscle strengthening of knee extensors	6-s contraction, 6-s relaxation. 3 sets of 10 repetitions
	Isotonic quadriceps exercise	Patient in the sitting position. Knee range of movement from 0º to 90º	Muscle strengthening of knee extensors	3 sets of 10 repetitions
	Isotonic hamstring exercise	Patient the sitting position. Knee range of movement from 0º to 90º	Muscle strengthening of knee flexors	3 sets of 10 repetitions
	Hip abduction and adduction	Patient in the long sitting position and knee extension, slight hip flexion to adduct, moving over the propped leg. Then the hip is abducted with the opposite movement. A load can be placed on the ankle depending on patient capability and progression	Muscle strengthening of hip abductors and adductors	3 min
Functional and balance	Ramps and steps	Patient climbs up by placing first the nonoperated leg and down by placing first the operated leg. Handrails are available for safety reasons	Weight bearing and resistance of lower limbs	3 min each
	Tandem position and walk	Patient keeps a position with the rear toe facing the heel of the front foot and stands keeping position. Then takes 4 steps backwards and 4 steps forwards	Standing and walking capabilities, coordination, static and dynamic balance	3 min each
	One leg standing position	Patient stands on one leg. Progress starting from weight-bearing changes with 2 legs to 1 leg standing, depending on patient progression. Handrails were available for safety reasons	Weight bearing and resistance. Standing and static balance	3 min
	Freeman and Bohler plates	Anteroposterior, mediolateral, and multiple direction movements placing the operated leg over the instruments	Improve proprioception and balance	3 min each

Appendix 2. Additional Training on Dynamometric Platform

Exercise	Description	Objective	Dose
Postural stability	Patient reaches markers placed on different parts of the instrument screen grid. The number of targets increased each week starting from 4 and reaching 6.	Improve proprioception, postural control, stability, and balance.	5 min
Weight-shifting	Patient shifts the weight from one leg to another in the medial-lateral, anterior-posterior, and diagonal planes	Weight bearing, postural control, and balance	5 min
Limits of stability	Patient proceeds through a movement pattern consistent with the area that can move the center of gravity within the base of support	Improve proprioception, postural control, stability, and balance	5 min

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